Phase 7 notes: comparing with optimization

Draft notes

1 Phase 7: Fourier stroke comparison

The goal of Phase 7 is to compare the discrete strokes discovered by PPO with smooth periodic strokes represented by a low-order temporal Fourier basis. This phase is motivated by the Phase 6 result that factored-action PPO with scaled radius produces robust coherent strokes at $N = 12$, including the refined $\Delta\theta = \pi/40$ sweep. Rather than further increasing N or refining $\Delta\theta$, we now ask whether the learned RL strokes are close to smooth flux-optimizing loops in the same shape variables.

We write the relative joint angles as

$$\phi_j(\tau) = a_{j,0} + \sum_{k=1}^{K} \left[a_{j,k} \cos(2\pi k\tau) + b_{j,k} \sin(2\pi k\tau) \right], \qquad \tau \in [0, 1],$$

for $j = 1, \ldots, N$. The first step is diagnostic: fit this representation to the PPO strokes and determine how many temporal modes are needed to reproduce the observed tip paths, cumulative spatial angles, and reward traces. The next step is direct optimization of the Fourier coefficients for total cycle flux, using the same geometry, hydrodynamic model, and wall/contact constraints as in the PPO environment.

This comparison separates three questions: whether PPO has found a smooth low-dimensional stroke family, whether that family is close to a classical flux-optimal loop, and how the flux-optimal stroke differs from strokes selected by efficiency or work-penalized objectives. Since the current PPO reward does not include an energetic cost, we expect PPO to be closer to a flux-optimizing stroke than to a true efficiency optimum.